

What choices do we have for 1 and 2 Qubit Gates?

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Quantum Gates

Pauli X

Gate	Description
Pauli X 	It rotates the qubit 180 degrees in the X-axis. Maps $ 0\rangle$ to $ 1\rangle$ and $ 1\rangle$ to $ 0\rangle$. Also it is known as the bit flip or NOT gate. It is represented by the matrix: $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Quantum Gates

Pauli Y

Pauli Y



It rotates around the Y-axis of the Bloch sphere by π radians.

It is represented by the Pauli matrix:

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

where $i = \sqrt{-1}$ is known as the imaginary unit.

Quantum Gates

Pauli Gates - Comparison

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Pauli Y 	It rotates around the Y-axis of the Bloch sphere by π radians. It is represented by the Pauli matrix: $Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ where $i = \sqrt{-1}$ is known as the imaginary unit.
Pauli Z 	It rotates around the Z-axis of the Bloch sphere by π radians. It is represented by the Pauli matrix: $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Quantum Gates

Hadamard

Hadamard



It represents a rotation of π on the axis $(X+Z)/\sqrt{2}$.

In other words, it maps the states:

- $|0\rangle$ to $(|0\rangle + |1\rangle) / \sqrt{2}$
- $|1\rangle$ to $(|0\rangle - |1\rangle) / \sqrt{2}$

This gate is required to make superpositions.

Quantum Gates

C-NOT

Controlled NOT (CNOT)



This is a 2-qubit gate that flips the target qubit (applies Pauli X) if the control is in state 1. This gate is required to generate entanglement.

Quantum Gates

S or Phase Gate

Phase $\sqrt{Z}$



It has the property that it maps $X \rightarrow Y$ and $Z \rightarrow Z$. This gate extends H to make complex superpositions.

Transposed conjugate of S



It maps $X \rightarrow -Y$ and $Z \rightarrow Z$.

Quantum Gates

T or Phase Gate

Gate	Description
Phase $\sqrt{S}$ 	The $\sqrt{S}$ gate performs halfway of a 2-qubit swap. It is universal such that any quantum multi-qubit gate can be constructed from only sqrt(swap) and single-qubit gates. It is represented by the matrix: $\sqrt{S} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2(1+i) & 1/2(1-i) & 0 \\ 0 & 1/2(1-i) & 1/2(1+i) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Transposed conjugate of T or T-dagger 	Represented by the matrix: $\sqrt{S}^\dagger = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2(1-i) & 1/2(1+i) & 0 \\ 0 & 1/2(1+i) & 1/2(1-i) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Quantum Gates

Measurement

Measurement



The measurement gate takes a qubit in a superposition of states as input and spits either a 0 or 1. Furthermore, the output is not random. There is a probability of a 0 or 1 as output which depends on the original state of the qubit.

Quantum Gates

If

Conditional

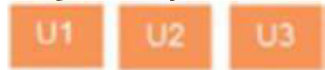


Conditionally apply a quantum operation.

Quantum Gate

U – Partial Rotations

Physical partial rotation (U gates)



U1: It is a one parameter single-qubit phase gate with zero duration.

U2: It is a two-parameter single-qubit gate with duration of one unit of gate time.

U3: It is a three-parameter single-qubit gate with duration of two units of gate time.

Quantum Gates

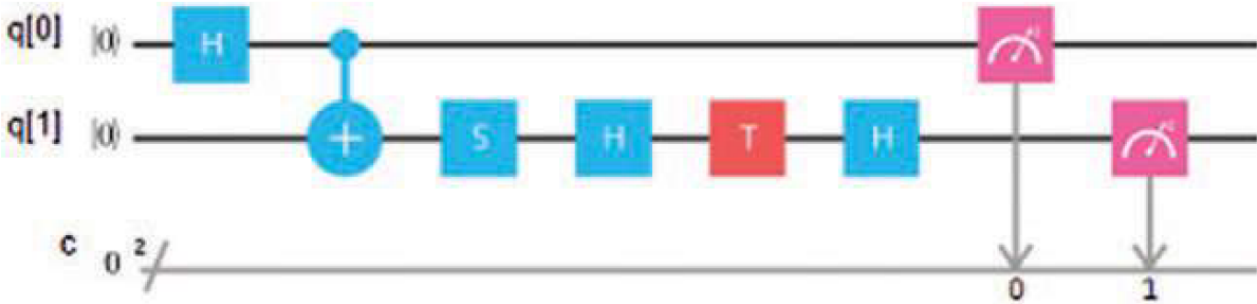
ID

Identity



The identity gate performs an idle operation on the qubit for a time equal to one unit of time.

Table 3-4. Quantum Circuits for Bell States

Bell state measurement		Result for 100 shots	
AB (ZW)			
	c[2]	Probability	
	11	0.39	
	10	0.06	
	00	0.46	
	01	0.09	

Classical Boolean logic

- What happens if we apply the gate to a quantum state?
- We permute the amplitudes!

$$X|\psi\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_0 \end{pmatrix}$$
$$X|\psi\rangle = X(c_0|0\rangle + c_1|1\rangle) = c_0|1\rangle + c_1|0\rangle$$

- Also called the "Pauli X" gate

$$|a\rangle \rightarrow |\neg a\rangle$$

Verify X is unitary.

Controlled-NOT gate

$$|a, b\rangle \rightarrow |a, a \oplus b\rangle \quad \text{If } a \text{ equals 1, then } b \text{ is flipped} \\ \Rightarrow b \text{ undergoes an } X \text{ gate}$$

Verify CX is unitary.

$$CX|\psi\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} c_0 \\ c_1 \\ c_3 \\ c_2 \end{pmatrix}$$

Example:

$$CX|+, 0\rangle = CX \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Verify this using matrix – vector multiplication.

Changing the phase

- How do we modify phase?

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$
$$Z(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle - \beta|1\rangle$$

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$
$$S(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle + i\beta|1\rangle$$

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$
$$Y(\alpha|0\rangle + \beta|1\rangle) = -i\beta|0\rangle + i\alpha|1\rangle$$

$$I = -iXYZ$$

$$XZ = -ZX$$

General single-qubit unitary gate

- A single-qubit gate can in general be written as

$$U = e^{i\gamma} V$$

$$V = \begin{pmatrix} \alpha & -\beta^* \\ \beta & \alpha^* \end{pmatrix}$$

Satisfies normalization condition: $|\alpha|^2 + |\beta|^2 = 1$

An important rotation gate

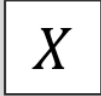
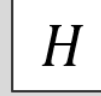
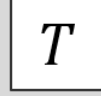
$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

We can view S and Z also as rotation gates.

$$e^{i\pi} = -1, \quad e^{\frac{i\pi}{2}} = i$$
$$Z = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{pmatrix}$$

$$T^2 = S, \quad S^2 = Z, \quad T^4 = Z$$

Quantum gates

Gate	Matrix	Symbol
Pauli X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	
Pauli Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	
Pauli Y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	
Phase S	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	
Hadamard H	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	
T = " $\pi/8$ "	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$	
Controlled Not CNOT	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	